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# 324551(24)

## B. E. (Fifth Semester) Examination, Nov.-Dec. 2016

**(New Scheme)**

**(Elect. Engg. Branch)**

**ELECTRICAL MACHINES-II**

***Time Allowed : Three hours***

***Maximum Marks : 80***

***Minimum Pass Marks : 28***

***Note :*** *Answer all questions. Part (a) from each unit is compulsory and carries 2 marks. Attempt any two parts from part (b), (c) and (d) of each question and carries 7 marks.*

### Unit-I

1. (a) Define field energy and coenergy. 2



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(b) Distinguish between singly-excited and doubly excited systems. For a singly-excited linear magnetic system, derive an expression for the electromagnetic torque. 7

(c) Explain the effect of harmonics on pitch factor and distribution factor. Prove that $\frac{E_{ph_n}}{E_{ph_1}} = \frac{KW_n\ B_n}{KW_1\ B_1}$

where $E_{ph_n}$, $KW_n$ and $B_n$ are the emf induced, winding factor and peak value of flux distribution in the airgap respectively for $n^{th}$ harmonic.

$E_{ph_1}$, $KW_1$ and $B_1$ are the same as above respectively for fundamental component. 7

(d) A 3-phase star connected alternator has 48 slots 4-poles and double layer winding with coil span equal to 10 slot pitches. The flux density distribution in the air gap is given by

$$B(\theta) = \sin\theta + 0\cdot4\sin 3\theta + 0\cdot25\sin 5\theta$$

Determine the rms value of 3rd and 5th harmonic phase emfs in terms of fundamental phase emf. Also



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the mechanical torque lost in friction be 13·5 N-m, find the copper loss in rotor windings, the input to the motor and efficiency. Stator loss is 750 watt. 7

(c) The rotor of a 3-phase induction motor has 0·04 Ω resistance per phase and 0·2 Ω stand still reactance per phase. What external resistance is required in the rotor circuit in order to get half of the maximum torque at starting. Neglect stator impedance. By what percentage will this external resistance will change the current and power factor at starting? 7

(d) Explain in detail with relevant circuit diagram, how the no-load test of an induction motor is performed? Also write the procedure to obtain the friction and windage loss from the no load test results. 7

## Unit-V

5. (a) Explain why a starter is necessary for starting a high capacity induction motor? 2

(b) Justify, a Star-Delta starter is equivalent to an auto-transformer/starter with 57·8 percent tappings. 7

(c) What is a double-cage induction motor? Draw the equivalent circuit of a double-cage induction motor.



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Sketch the torque-speed characteristics of double cage induction motor. 7

(d) A 440 V, 3-phase, 4-pole 50 Hz squirrel cage induction motor develops a maximum torque of 3 times the full load torque at a speed of 1200 rpm and at rated voltage and frequency. If the motor is now connected to 400 V, 40 Hz ac source, then neglecting the stator resistance effect, compute : 7

(i) the magnitude of maximum torque.

(ii) the speed of which the maximum torque would now occur.



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